ACE MANUFACTURING SYSTEMS LTD.,.

PACKING LIST

Machine No: M35-001

Customer: M/s. SUNDARAM CLAYTON LIMITED., FOUNDRY DIVISION CHENNAI.

Reference: Your Purchase Order No. **Dated:**

Sl No.	Description
1	**CNC Vertical Machining Center Model "MCV-650"**
	Spindle nose taper 7/24 No.40
	Spindle drive (Fanuc) 30 Min./Cont. Rating - 11 / 7.5 KW
	AC Servo axes drive (Fanuc)
	Twin Arm (Random) type auto tool changer
	Tool shank BT-40
	LM Guideways for X,Y & Z Axis
	CNC System Fanuc 0IM with 7.2" Monochrome LCD
	Graphics - Single colour
	Ball screw stretching for X, Y & Z Axes
	Panel cooler for electronics
	Counter balancing for Z-Axis
	Fail safe brake for Z-Axis
	Flexible coupling for X, Y & Z Axes
	Standard work light
	Rigid tapping
	Total machine enclosure
	Chip disposal to the rear side of the machine
	Spindle speed of 80 - 8000 Rpm (Direct drive)
	Air drier unit (Small)
2.0	High pressure coolant through spindle system (16 lpm at 21 bar) with Grundfoss centrifugal pump and bag type filtration system (with hollow shaft motor)
3.0	Flood coolant system around the spindle
4.0	Chip flushing system on the base
5.0	Coolant gun
6.0	Special colour
7.0	Screw conveyor (2 Nos.)
8.0	Special guarding with 3 auto doors
9.0	24 station tool magazine in lieu of 20 station tool magazine
10.0	Scraper type chip conveyor with drum type filtration for cast iron & aluminium chips (Conveyor position parallel to X-axis)

11.0	Tool life management
12.0	TPM points (see through cabinet, markings and other TPM points as per SCL requirement)
13.0	Hydraulic power pack for the fixture (with 4 station manifold)
14.0	Modifications to the hydraulic power pack to enable high low clamping arrangement for SCL supplied fixture for enabling hydraulic jack
15.0	Fixture interfacing along with necessary encodes (fixture is not in our scope)
16.0	Additional operation panel with single lever for cycle start & stop
17.0	Manual pulse generator (Handheld)
18.0	Three tier indication lamp with hooter
19.0	Machine lamps - 2 Nos.
20.0	Manuals - 2 sets
21.0	Following reports to be sent along with the machine.
	1. Geometrical test
	2. NASA test
	3. Laser calibration test
	4. Heavy cut & Rigid tapping test

AMS	ACCEPTED UNDER DEVIATION	DATE: 22/09/06

TO: ASSEMBLY DEPARTMENT

FROM: VENDOR DEPARTMENT

SUB: Deviation in Column.

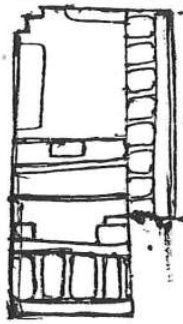
DESCRIPTION: Structure Column

DRAWING NO: C87-0128-2540. PART SL.NO.:

MACHINE NO: M35-001.

DETAILS :

Guide way length $1380^{\pm 0.2}$ mm is measured 1375 mm. Rough m/cing mistake shown here.



Actual is 1375 mm

ACTION PLAN :

The actual Bond diff between LM guide end & column End face to be measured & the diffrn in to be compensated by grinding / milling in ~~BKT~~ supporting ~~BKT~~ face



DEVIATED SOURCE: Hampa Industries.

AMOUNT TO BE DEBITED TO: Hampa Industries.

INSPECTED BY: Lakshmisha.H.

INSPECTION REPORT NO:

DEVIATION ACCEPTED BY:	DESIGN	ASSLY	APPROVED BY	DATE:

COPY TO - SUPPLIER / CASE FILE / STORES.

RP / NS / KM / MR / LS / QUALITY

FORMAT REVISION 11.08.2006

21/9

FORMAT

AUD

SM. IR: FOR N.A PLS.

Sep30, 2006

Minutes of Meeting held between M/s.M/s. Ace Manufacturing Systems Ltd., Bangalore & Ms.Sundaram Clayton Ltd., Chennai (Die cast Division) at AMS works on 30.9.2006.

AMS
Mr. S.Ananthakrishnan
Mr. Muralidhar Rao
Mr. Keshava Murthy
Mr. Madhukar

SCL
Mr.K.Ravi Kumar

Sub: Inspection and Dispatch clearance of 2 nos. MCV-650 & 1no.HMC-400XL

Mr. K.Ravi Kumar visited AMS on 29th, & 30th Sept 06 for Inspection of 2 nos. MCV-650 & 1no.HMC-400XL

The following points to be attended before dispatch:

MCV-650 (M35- 001& 002)

1. Heavy cut and NASA test to be done
2. Circularity test for Dia 600 mm to be done.
3. Screw conveyor to be fixed.
4. ATC guard to accommodate Dia 205 boring bar.
5. ATC pockets to be protected from entry of chips/ burrs.
6. Name plate for pressure indicator of Pneumatic counter balance.
7. Check Z-axis drop during power failure.
8. Two hand cycle start required in addition to strike switch for cycle start..
9. Chip trolly required.

HMC-400XL :

1. Additional push button required for right side door in the pendent.
2. Geometrical inspection of the machine should be carried with false plate also.
3. Hardness of Table & spindle collet to be informed to SCL.
4. Index table Rotary joint to be checked with 3 lines @ 70 bar AND 4bar Pneumatic pressure to see there is no leakage / mix up of oil & Air.
5. Spindle collet open/ close push button to be provided on the side of the pendent with cover.
6. 300 mm tool length should be accommodated in the Magazine.
7. Front Auto door to be >900 mm.
8. Identification required for the Rotary joint ports wrt Power pack.
9. Pneumatic counter balance adjustment procedure to be incorporated in the manual.

General Points:

1. CTS rubber seal on spindle face should not fall if CTS is on without tool holder- *AMS will study* and revert back.
2. Spindle should not rotate without tool - *AMS will study and revert back.*
3. ATC arm getting struck frequently and Tool pot nylon pads are getting loosened often. Some times tool also falls down – *AMS will ensure reliability of ATC functions.*
4. Spindle Star contactor not on ALARM is frequently coming on recently supplied MCV-600 & MCV-600CX machines.- *AMS will find a solution for this alarm.*

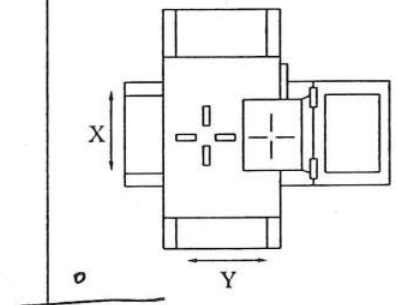
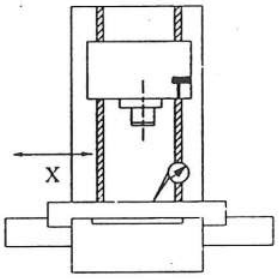
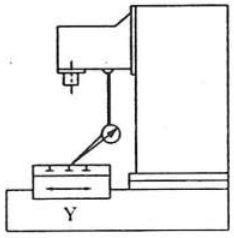
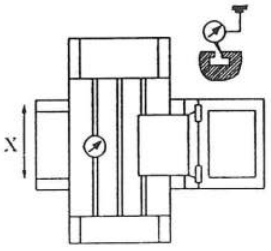
AMS to attend to the points and dispatch the machine on 3rd Oct-06.
AMS thanks SCL for deputing Mr.Ravi Kumar for inspection & clearance of the machine..

For AMS **For SCL**

M35-001

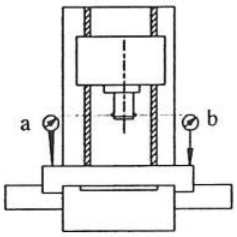
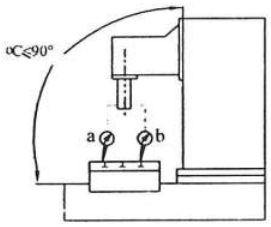
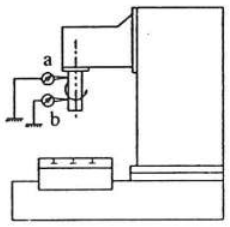
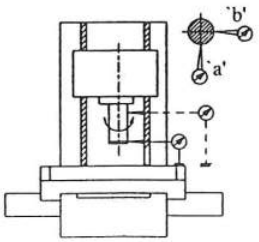
MCV 650

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-600H

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G1	Levelling of Machine X & Y axes at their mid stroke -10		0.04/m along X & Y axes.	Level along X-axis : Level along Y-axis :
G2	Parallelism of pallet surface w.r.t. its Guideways or X-axis movement +4 -25 0 -8.5 -11 0 @Y350 +1 -11 0 @Y650 ±12		0.04/1000	0.015
G3	Parallelism of pallet surface w.r.t. Y-axis Guideways or Y-axis movement -22 -21 -6 -5 -18 -25 -10 0 0 0 0 2μ 1μ		0.025/600	0.022
G4	Parallelism of reference T-slot to X-axis movement 0 -8 -20		0.04/1000	Pallet A 0.020 Pallet B

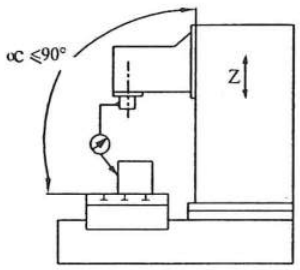
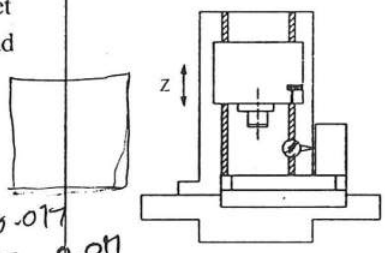
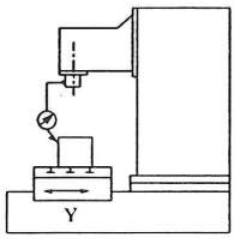
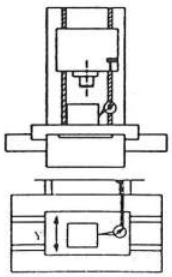
DATE : 30/09/08	CHD. CUST. : V. Rawat	AMD. DT. :	DOC. NO. D23-001-01-03	
BY :	CHD. AMS :	SHEET : 01 OF 04	PAGE NO.	F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-600H

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	Squareness of the Spindle axis to the pallet surface in the XZ plane. X & Y axes at their mid stroke		0.020 for 300 swing diameter ϕ430	0.014 – 29/09 0.015 – 30/09
G6	Squareness of the Spindle axis to the pallet surface in the YZ plane. X & Y axes at their mid stroke		0.02 for 300 swing diameter ∝C≤90° ϕ430	0.010 – 29/09 0.009 – 30/09
G7	Run out of spindle taper a) Near the Spindle nose b) At a distance of 300 from the spindle nose		a) 0.01 near the spindle nose b) 0.02 at a dist. of 300mm.	a. 0.002 b. 0.01
G8	Parallelism of spindle Axis & Z-axis in a) XZ plane b) YZ plane to be checked at mean point of runout		a) 0.020/300 b) 0.020/300	0.027 29/09 0.004 30/09 0.004 0.016 0.009

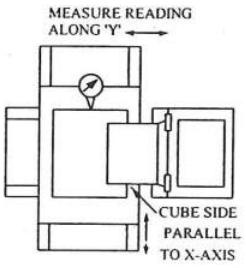
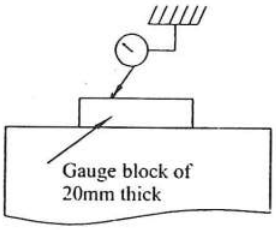
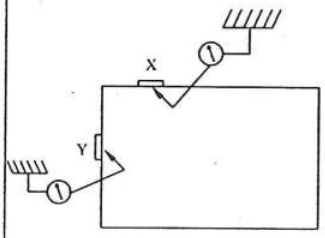
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BY :	CHD. AMS :	SHEET : 02 OF 04	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-600H

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G9	Squareness between pallet surface & the spindle head along the Z-axis in XZ plane +0.15 3 +0.20 +12 0		0.03/450	0.015
G10	Squareness between pallet surface & the spindle head along the Z-axis in YZ plane 0 0.028 0.017 0.017		0.03/450	-0.028 correct -0.017
G11	Straightness of X-axis movement in XZ plane		0.01/500	0.006
G12	Straightness of Y-axis movement in YZ plane		0.01/500	

DATE : 30/9/06	CHD. CUST. : V. Ravi	AMD. DT. :	DOC. NO. D23-001-01-03
BY :	CHD. AMS :	SHEET : 03 OF 04	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-600H

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G13	Squareness of Y-axis movement w.r.t. X-axis movement		0.023/450	0.038 X 0.012 After correcting taper
G14	Variation between the pallet height		0.03	
G15	Pallet repeatability in X & Y axes when loaded on the machine from the pallet station for same pallet		0.02 in X & Y axes direction	
	Axial play Radial play	0.003 0.005		

DATE : 30/9/06	CHD. CUST. : V. Raut	AMD. DT. :	DOC. NO. D23-001-01-03
BY :	CHD. AMS :	SHEET : 04 OF 04	PAGE NO. F3

CHECK POINTS		
1	Ensure plugs are fitted for L M Guideways	
2	Ensure effective wiping of saddle cover wipers	
3	Check the tightness of coupling and is set to 17 Nm Torque	
4	Ensure lubrication presure switch is set to to 15 bar	
5	Ensure lub timer is set 12 min for H G and 20 min for LM guideways.	
6	Check the belt tension by using belt tension meter	
7	Ensure the proper supply of air to spindle at oriented condition.	
8	Ensure air presure switch is working properly.	1 2
9	Ensure proper setting of oil in F R L unit	
10	Check the tightness of push stud.	
11	Check oil in ATC gearbox.	
12	Ensure grub screws are fixed to locknuts in ATC.	
13	Ensure grub screws are fitted to spindle flange.	
14	Ensure sliding door has free movement without play and noise.	
15	Ensure no coolant leakage around the machine.	✓
16	Ensure drum wash in conveyors.	
17	Ensure proper working of float switch.	✓
18	Ensure levelling bolts are fixed with locknut..	
19	Check Table Tenon Slot 14H8/18H8 [Tolerance_0 to +0.027mm]	
20	Ensure air blast in spindle after CTS is switched off.	✓
21	Ensure identification name plates are fixed in power pack hoses.	
22	Ensure bag filter clogging.	✓
23	Check spindle orienation position and note down the value.4031 --	
24	Ensure suction strainer in CTS tank	
25	Check conveyor overload message .	✓
26	Check pallet Intilation in APC machine	-N·A-
27	Check spindle balancing.	
28	Ensure the three axes Ballbar Completion	
29		

[Mechanical check list]

*Due to urgent m/c dispatch Machine Inspection not done

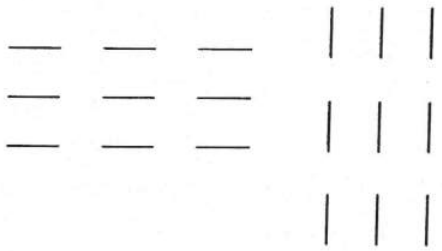
* Machine despatched without work-orders.

Kulal.
30/9/2006.

GEOMETRICAL TEST CHART

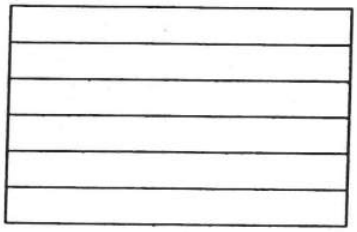
DATE :

1. SPRIT LEVEL READING

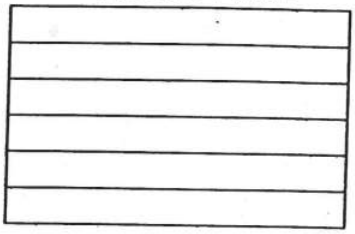


2. TABLE TOP READINGS

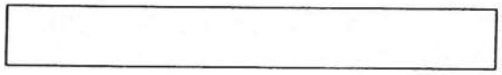
ALONG X-AXIS



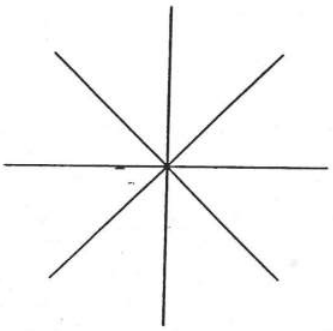
ALONG Y-AXIS



3. T SLOT READING

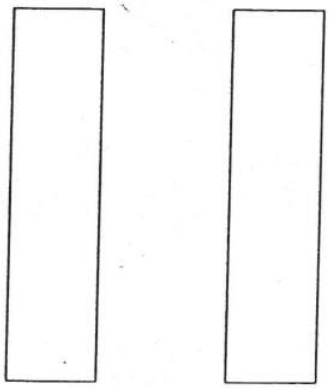


4. TURN OVER READING



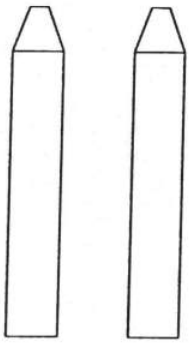
5. COLUMN READING

F S



6. MANDREL READING

F S



RUN OUT:

a. TOP

b. BOTTOM

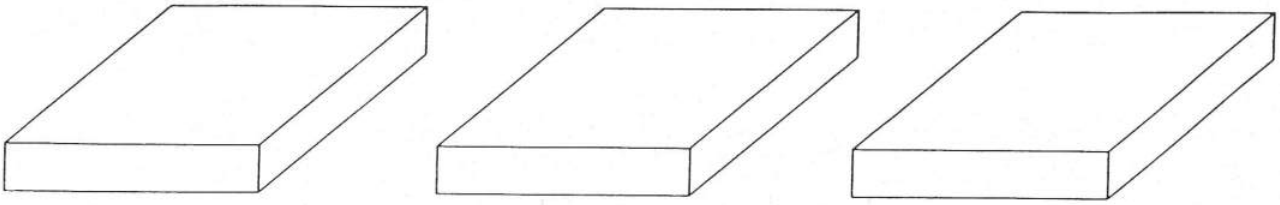
(I)

(II)

(III)

(IV)

7. SQUARENESS READING (X & Y)



CUSTOMER'S NAME:		DATE :		
MACHINE No:				
CHECK LIST				
BEFORE DISPATCHING OF THE MACHINE THE FOLLOWING POINTS TO BE CHECKED.				
SL No.	**PARTICULARS**			
I	**AXIS :**			
1	Wheather the limit switch for all dog setting compressed only by 2 mm for X and Y axis , 3.5 mm for Z axis. **Note : Not applicable for β - motors.**	X	Y	Z
		NA		
2	Check the marker pulse of all the axis are: **Note : Not applicable for β - motors.**	X	Y	Z
		NA		
3	Check the backlash of all the axis in MPG mode	X	Y	Z
	PARAMETER 1851			
	PARAMETER 1852			
4	Check the Z - axis motor current during Tool de - clamping :			
	a - WINNER / MCV 320 - 130 to 150% -	840 Feed rate in jog mode -		
	b - DTC - / 110 to 130% -	Auto Cycle -		
5	Z - axis drop during power off :			
	a . + ve direction (Towards table) -			
	b . - ve direction -			
	SYSTEM--			
	DIAL----			
II	**AXES**			
1.a	Percentage of axis current drawn :			
Axis	25% Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm Feed Rate
X				
Y				
Z-Down				
Z - Up				
1.b	Percentage of Axis holding current:			
Axis	100 % Rapid rate		2000 mm Feed Rate	
	Motor End	Brake End	Motor End	Brake End
X				
Y				
Z				
1.c	Over travel against Axis dead stopper in mm:			
Sl No.	Axis	Motor End	Brake End	
1	X			
2	Y			
3	Z			

III	SPINDLE :			
1	Record the distance of push stud face from the spindle face when there is no tool :			
	a . Tool ejection :	**Allowable**	**Auto**	**Manual**
	BT - 40	**0.7/1.0**		
	BT - 30	**0.5**		
	b. Tool Clamping Force (Kgf) -			
2	Ensure that the gap between draw bar and ejection screw / Roller			
	a . When there is no tool in the spindle (1 mm):			
	b . When tool holder in the spindle (3 mm) :			
3	Axial play in the spindle(mm) :			
	a . Tool De - Clamping :	O → →	**Difference**	
	b . 130 % Load by wood :	O → →	= mm	
4	Percentage Current drawn at maximum R.P.M of the spindle:			
5	**SPINDLE STABILISATION**			
SL No.	**TIME**	**SPINDLE TEMP**		
1				
2				
3				
4				
5				
	FINAL ROOM TEMPERATURE :			
	FINAL SPINDLE STABILISATION TEMPERATURE :			
IV	**A.T.C :**			
1	Check the spring lever compression (mm) :			
	a . Z - axis below to zero position			
	b . After the magazine buts against the dead stopper			
	c . Difference - Winner (2 to 3 mm) / DTC (1 to 2 mm).			
2	Tool picking position in mm	Actual	9001 Program	
3	Back Lash in the A.T.C disc gear(mm) :			
V	**A.P.C**			
1	Pressure setting for the pallet transfer cylinder in kg/cm2 :			
2	Pressure setting for the pallet Clamp - Declamp in kg/cm2:			
VI	**CYCLE TIME :**			
1	Check the chip to chip time Sec			
2	Check the tool to tool time Sec			
3	Check the palet changing cycle time Sec			
VII		X	Y	Z
1	Time constant linear- 1620			
2	Time constant bell-1621			
3	Acceleration time in sec:			
4	Decceleration time in sec:			

AMS	MACHINE HISTORY			
1	**STANDARD MACHINE:**			
		X -AXIS	Y -AXIS	Z -AXIS
1.1	BALL SCREW PITCH mm			
1.2	DISTANCE FROM SPINDLE FACE TO TABLE TOP WITH INCREASED DAY LIGHT AREA	mm.		
2	**FLEXIBLE COUPLING**	X - AXIS	Y - AXIS	Z - AXIS
2.1	**MODEL**			
2.2	**MAKE**			
3	**ATC DETAILS:**			
3.1	ATC (TURRET TYPE / PRAGATI)	SL.NO		
3.2	TOOL CAPACITY	NOS.		
3.3	SIDE CAM ROLLER	MAKE		
		MODEL		
		MAKE	**MODEL**	**SL No.**
3.4	ATC MAGAZINE ENCODER			
3.5	ATC BRAKE	EMCO		
3.6	ATC ARM MOTOR BRAKE			
3.7	ATC TORQUE MOTOR	TM 50 /TM 80 /TM 100 /TM 125		
4	**AXES MOTORS**	X - AXIS	Y - AXIS	Z - AXIS
4.1	MODEL			
4.2	MAKE	FANUC	FANUC	FANUC
4.3	SL No.			
4.4	TYPE			
5	**TELESCOPIC COVERS**	DWG No.		MAKE
	X -AXIS			SUR
	Y -AXIS			HENN
	Z - AXIS			ING
6	**BRAKES**			
	MAKE	X - AXIS	Y - AXIS	Z - AXIS
		NA	NA	In Built
7	**HEAD STOCK DETAILS:**			
7.1	SPINDLE DETAILS :	STD / CTS		
7.2	SPINDLE MOTOR	MAKE	SPECIFICATION	
		FANUC		
7.3	SPINDLE NOSE TAPER	BT		
7.4	CARTRIDGE No.			
7.5	SPINDLE SPEED	RPM		
7.6	SPINDLE BEARINGS Dia			
7.7	SPINDLE DIRECT DRIVE / INDIRECT DRIVE			
7.8	PULLEY RATIO			
8	**BELT DETAILS: NOT APPLICABLE TO DIRECT DRIVE.**			
		GT BELT	**COUPLING**	**REMARKS**
8.1	**MODEL**			
8.2	**MAKE**			

AMS	MACHINE HISTORY			
9	**COOLANT SYSTEM DETAILS**			
9.1	TANK CAPACITY (LTRS) :	300		
	RAJMANE INDUSTRIES	**MODEL**	**SL No.**	**TYPE**
9.2	FLOOD COOLANT PUMP			CENTRIFUGAL PUMP
9.3	CHIP PUSHING COOLANT PUMP			
9.4	BASE FLUSHING COOLANT PUMP			
9.5	COOLANT GUN & DRUM WASH PUMP			
9.6	HIGH PRESSURE COOLANT PUMP	MAKE		
		MODEL		
9.7	PRESSURE RELIEF VALVE	MAKE		
		MODEL		
9.8	ROTARY JOINT	MAKE		
		MODEL		
10	**FLOAT SWITCH**			
10.1	MAKE	FILPRO ELECTRONICS		
10.2	PART NO			
11	**CONVEYOR DETAILS**			
11.1	CONVEYOR DETAILS			
	TYPE OF CONVEYOR			
	MODEL			
	MAKE	UNIVERSAL CONYEORS		
	SL No.			
11.2	**GEAR BOX DETAILS :**			
	MODEL			
	MAKE	UNIVERSAL CONYEORS		
	SL No.			
11.3	**FILTER SPECIFICATIONS :**			
	MODEL			
	MAKE	UNIVERSAL PRECEISION TOOLS		
	SL No.			
11.4	**COIL / SCREW CONVEYOR DETAILS**			
	MODEL			
	MAKE	UNIVERSAL PRECEISION TOOLS		
12	**HYDRAULIC SYSTEM**			
12.1	POWER PACK			
	PRODUCT No.	Q.C N0-		
	TANK CAPACITY (LTRS)			
	MAXIMUM WORKING PRESURE			
	MAKE			
12.2	PRESSURE SWITCHES			
	OPERATING RANGE			
	MODEL			
	MAKE	POLY HYDRON		
	QUANTITY			

AMS	MACHINE HISTORY			
12.3	ACCUMULATOR			
	MAKE			
	SET OIL PRESSURE			
	GAS PRESSURE			
	MODEL			
12.4	**HYDRAULIC CYLINDERS:**			
	MAKE	**MODEL**	**FUNCTION**	**QTY**
1	ISLAND HYDRAULICS		PALLET F/R	2
2				
12.5	**HYDRAULICS VALVES**			
	TYPE	**MAKE**	**MODEL**	**QTY**
1	DIRECTIONAL VALVE	YUKEN	DSG-01-3C4-D24-N1-50	
2	REDUSING MODULAR VALVE	YUKEN	MRP-01-B30	
3	PILOT OPERATED CHECK VALVE	YUKEN	MPW-01-2-30	
4	THROTTLE & CHECK MODULAR VALVE	YUKEN	MSW-01-X-30	
5				
6				
12.6	HYDRAULIC PUMP	MAKE	MODEL	TYPE
12.7	MOTOR	MAKE	SL No.	MODEL
		SIEMENS		
13		MAKE	SL No.	PRESSURE SETTING
	LUBRICATION UNIT DETAILS			15Kg/cm2
	MOTOR			
14	**FRL UNIT**			
	a . MAKE	SHAVO		
	b . MODEL	B-13-296-M6LC/L-13-200-MP8C		
	c . OPERATING RANGE	0-10Kg/cm2		
	d . GRADE OF OIL	SERVO WAY SPIN 12		
14.1	DRYER	MAKE	SPECIFICATION	
		TRIDENT	VXD-2	
15	**PNEUMATIC VALVES**			
SL No.	**APPLICATION**	**SPECS**	**MAKE**	**QUANTITY**
1	SPINDLE AIR JET	MFH 3-1/8	FESTO	1
2	PNEUMATIC COUNTER BALANCING	DC-220-3E1-DC 24	SHAVO	1
3	MANUAL TOOL DE-CLAMP	MFH 5-1/8	FESTO	1
4	ATC POCKET UP&DOWN/APC AUTO DOOR	JMFH 5-1/8	FESTO	
5	CTS AIR JET	MFH 3-1/8	FESTO	
6	FRONT AUTO DOOR	JMFH 5-1/8	FESTO	
		MFH3-1/8	FESTO	
7	PALLET REST PAD CLEANING	MFH 3-1/8	FESTO	
8	FIXTURE		FESTO	

AMS	MACHINE HISTORY			
15.1	**PRESSURE SWITCHES:**			
	OPERATING RANGE	2 TO 15 Kg/cm2		
	MAKE	SUCO		
	QUANTITY	1+		
15.2	**PNEUMATIC CYLINDERS USED (BOUGHT OUT - IF ANY)**			
SL No.	**FUNCTION**	**MAKE**	**MODEL**	**QTY**
1	PNEUMATIC COUNTER BALANCING	FESTO	DNU- -600-PPV	1
2	MANUAL TOOL DE-CLAMP	FESTO	DNU - PPV	1
3	FRONT AUTO DOOR	FESTO		
4	APC AUTO DOOR	FESTO		1
16	**TURRET / INDEXING TABLE (OPTIONAL)**			
	MAKE	**MODEL**	**SL No.**	**No. OF STATIONS**
	PRAGATHI			
17	**ROTARY TABLE (OPTIONAL) -**	MAKE	MODEL	SL No.
18	**LIMIT SWITCH:**			
SL No.	**MAKE**	**Qty**	**FUNCTION**	
1		1	FRONT AUTO DOOR	
2		2	APC AUTO DOOR	
3		2	PALLET IN STATION	
4				

AMS

ELECTRICAL MACHINE HISTORY

MACHINE NO : M35-001

DATE : 02/10/06

ACE MANUFACTURING SYSTEMS LTD,

PLOT NO.467 to 469,
4TH PHASE, PEENYA INDUSTRIAL AREA,
BANGALORE – 560 058. INDIA
TEL : 91-80-28360508 to 12
FAX : 91-80-28362791
Email : helpdesk@amslindia.co.in
Website : http://www.amslindia.com

KEEP RELAY AND OTHER DATAS

KEEP RELAY NOS (=1)	DESCRIPTION	
EX: K10.0	STD CLNT OPTION	
K1.5		
K2.0		
K3.4		
K3.6		
K5.0		
K6.1		
K10.1		
K10.2		
K10.6		
K11.3		
K12.3		
K12.4		
K12.5		
K12.6		
K16.4		
K16.5		
K17.1		
K3.3		
K24.3		
K37.0		
K39.0		
K39.5		
K40.0		

AXES	ACCELARATION		LOOP GAIN
EX: X	(PAR 1620 [T1] + PAR 1621 [T2]) 100 + 25		EX: (PAR 1825) 4000
X	80	64	8000
Y	80	64	8000
Z	120	120	8000
B			
CHIP TO CHIP TIME			
TOOL TO TOOL TIME:			

NAME: Prakash Bhat
SIGN:

	NOTE: CHECK THE FOLLOWING PARAMETERS SETTING IN ALL FUTURE MACHINES
1	Set the parameter 3402#6=1 (CLR) ✓
2	Spindle position feedback display in dgn 442.i.e. 3117.1 = 1. ✓
3	Graphics display enable 3112#0 =0 ✓
4	operator message history display. i.e. 3112.2 = 1. ✓
5	Alarm discription display in alarm history screen for external alarms.i.e. 3112.3 = 1. ✓
6	Operator history enable 3112.5=0 ✓
7	system alarm history display. i.e. 3103.2 = 1 ✓
8	Operation history enable. i.e. 3106.4 = 1. ✓
9	In the offset screen offset value CLEAR by softkey to be disabled. i.e. 3205.4 = 1. ✓
10	I / O channel setting should be made 0 before dispatching the machine ✓
11	over travel before move 1301#7=1 ✓
12	400 registered programme numbers 9935#6=1 * new system
13	over travel before move enable(along with 1301.7=1) 9964#3=1 * new system.
14	2062- Parameter over current ✓
15	3111 - Message ✓
16	1320 - Over Travel ✓
1.	Title editing as per customer with time and date ✓
18	Lubrication timer LM guideways: ✓ harden guideways : 120,000
19	Counter for disc spring life over : ✓
20	9000 series program locking ✓
21	Servo Guide backup & RS232 communication checking ✓
NAME:-	Prakash Bhat DATE:- 03/10/06
SIGNATURE:-	

MACHINE HISTORY

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

MACHINE MODEL	SPARK,DTC,MCV300,MCV320,WINNER,MCV400 ,MCV400S,MCV400SX	
	MCV500S,MCV-600CX,DART,DART-PLUS,MCV500 ,MCV600 ,HMC400	
	MCV-650 WINNER-XL,CHALLENGER,CHALLENGER PLUS,CMH	
MACHINE NUMBER	M35-001	
CUSTOMER NAME	SCL Chenai	
SRAM CAPACITY	1MB / 0.5MB	
SYSTEM	[FANUC / SIEMENS]	0I-MB / 0I-MC / 0I-MATE MC
MAIN OPTIONS	STDCLNT,C.CONV(SLAT/SCRAPER),D.CONV, CTS,HYFIX,	
	PNFIX,AUTODOOR,ROT.TB,SRVATC,SRVMAG,APC,I.TABLE	
DOCUMENT VERSION		

M/C BACKUP (PATH) - SERVER:

M/C BACKUP (PATH) : \\AMS37\D:\ELECTRICALLMACHINEDATA\AMSMACHINEDISPATCHBACKUP(16.08.2004)

FANUC SYSTEM	**TYPE NUMBER**	**SERIAL NUMBER**
0I - SYSTEM	A02B-0280-B502	
0I-MB SYSTEM	A02B-0299-B802	
0I-MC SYSTEM	A02B-0309-B500	E064B 2770
0I-MATE MC SYSTEM	A02B-0311-B500	
FANUC I/O UNIT	A02B-0309-C001	E07532 2006-04
FANUC I/O CARD WITH MPG CONN	A20B-2002-0520 / 07A	
FANUC I/O CARD WITHOUT MPG CONN	A20B-2002-0521 / 07A	
DISPLAY	CRT LCD (COLOUR/MONO)	
SVPM 5.5i (3 AXES + 1 SPNDL)	A06B-6134-H301#A	
SVPM 15i (3 AXES + 1 SPNDL)	A06B-6134-H303#A	
POWER SUPPLY MODULE		
PSMR 5.5(ALPHA-C3 & ALPHA-3 MTR)	A06B-6081-H106	
PSM11(ALPHA-C6 & ALPHA-6 MTR)	A06B-6077-H111	
PSM15 (ALPHA-8 & ALPHA 22/22/22B)	A06B-6087-H115	
PSM-26(ALPHA-8\A12 & ALL AXIS AC22)	A06B-6087-H126	
PSM11i(FOR SPαC6/6Ki / SPα8/10Ki MOTOR)	A06B-6110-H011	
PSMR5.5i(FOR SPα3/10Ki / SPαC3/6Ki MOTOR)	A06B-6115-H006	
PSM-26i (FOR A12/10Ki)	A06B-6110-H026	V0624 2297
ITEM	**TYPE NUMBER**	**SERIAL NUMBER**
SPINDLE AMPLIFIER MODULE		
SPMC-5.5(ALPHA-C3&ALPHA-C3I MTR)	A06B - 6082 - H206 - # H512	
SPM-5.5 (ALPHA -3 MTR)	A06B - 6082 - H206 - # H520	
SPM-5.5 (ALPHA -3I MTR)	A06B - 6102 - H206 - # H520	
SPMC 11(ALPHA-C6 &ALPHA-C6I MTR)	A06B - 6082 - H211 - #H512	
SPM -11(ALPHA-6 & 8 MTR)	A06B - 6082 - H211 - #H520	
SPM - 11(ALPHA-6I & 8I MTR)	A06B - 6102 - H211 - #H520	
SPMC-5.5i(FOR SPαC3/6Ki MOTOR)	A06B-6116-H006#H560	

MACHINE HISTORY

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

SPM-5.5i (FOR SPα3/10Ki MOTOR)	A06B-6111-H006#H550	
SPMC-11i (FOR SPαC6/6Ki MOTOR)	A06B-6116-H011#H560	
SPM-11i(FOR SPα6/10Ki / SPα8/10Ki MOTOR)	A06B-6111-H011#H550	
SPM-15i (FOR SPα12/10Ki) ✓T8	A06B-6111-H015#H550	V06419432
SERVO AMPLIFIER MODULE		
SVM3-20/20/20-B	A06B-6080-H304	
SVM1-40L	A06B-6079-H104	
SVM2-40/40	A06B-6079-H206	
SVM3-20/20/20i	A06B-6114-H303	
SVM1-40i	A06B-6114-H104	V06300920
SVM2-40/40i	A06B-6114-H207	V06226879
SERVO AMPLIFIER MODULE (B-AXES)		
SVU-20 (FOR A2/5Ki MTR.FOR 0i-MB SYSTEM)	A06B-6093-H112	
SVM1-20 (FOR ß3/3K,ß6/2K (For 0i-MA System))	A06B-6079-H102	
SVU1-20 (FOR ß3/3K,ß6/2K (For 0i-MA System))	A06B-6089-H102	
SVU1-40 (FOR A3/3K MOTOR)	A06B-6089-H104	
BSVM1-20i (FOR ß4/4Kis (For 0i-MB System))	A06B-6130-H002	
SERVO ARM /MAGZINE DRIVE		
B I/O LINK (FOR ß8/3Kis)[240V 6.3A,O/P 5.9A]	A06B-6093-H152	
BSVM1-20i (FOR ß4/4Kis)[240V 8A ,O/P 6.8A]	A06B-6130-H002	
(For 0i-MB System))		
ITEM	**TYPE NUMBER**	**SERIAL NUMBER**
SERVO ARM /MAGZINE MOTOR (PMC)		
B4/4Kis (0.75kw,4.7A,Ts-3.5Nm)	A06B-0063-B003	
B8/3Kis (1.2kw,153v,4.9A,Ts-7Nm)	A06B-0075-B003	
AXES MOTORS		
AC8/2Ki (1.2KW,5.6A,Ts-8Nm)	A06B-0226-B001 (X \ Y \ Z)	
AC12/2Ki (1.8KW,6.5A,Ts-12Nm)	A06B-0241-B101 (X \ Y \ Z)	
AC12B/2Ki (1.8KW,6.5A,Ts-12Nm)	A06B-0241-B401 (X \ Y \ Z)	
AC22/2Ki (3KW,12.3A,Ts-12Nm)	A06B-0246-B101 (X \ Y \ Z)	
AC22B/2Ki (3KW,12.3A,Ts-12Nm)	A06B-0246-B401 (X \ Y \ Z)	
B4/4Kis (0.75kw,4.7A,Ts-3.5Nm)	A06B-0063-B003 (X \ Y \ Z)	
B8/3Kis (1.2 KW,153V,4.9A,133HZ) STAR	A06B-0075-B003 (X \ Y \ Z)	
B8B/3Kis (1.2 KW,153V,4.9A,133HZ) STAR	A06B-0075-B (X \ Y \ Z)	
B12/3Kis (1.8 KW,138V,8.0A,133HZ) STAR	A06B-0078-B103 (X \ Y \ Z)	
B12B/3Kis (1.8 KW,138V,8.0A,133HZ)STAR	A06B-0078-B403 (X \ Y \ Z)	
B22/2Kis (2.5KW,11.3A)	A06B-0085-B103 (X \ Y [illegible]) ✓	C065B3748/C065B3539
B22B/2Kis (2.5KW,11.3A)	A06B-0085-B403 ([illegible] \ Z) ✓	C065B4012
4TH AXIS MOTOR		
A06B-0063-B003	ß4/4Kis(0.75kw,4.7A,Ts-3.5Nm)	
A06B-0075-B003	B8/3Kis (1.2 KW,153V,4.9A,133HZ) STAR	

MACHINE HISTORY

Date:-

NOTE:- ***CHECK THE MACHINE AS PER THE WORK ORDER***

SPINDLE MOTOR		
SPA6/12KiT (5.5KW/3.7KW,37-48A,111-220V)	A06B-1466-B123#0021	
SPAC3/6Ki (5.5/3.7 KW,O/P TORQUE=23.5NM)	A06B-1435-B100	
SPA3/10Ki (5.5/3.7 KW,29A,TORQUE=23.5)	A06B-1405-B103	
SPAC6/6Ki (7.5/5.5 KW,O/P TORQUE=35)	A06B-1436-B100	
SPA6/10Ki (7.5/5.5 KW,O/P TORQUE=35)	A06B-1406-B103	
SPA8/10Ki (11/7.5 KW,44-56A,TORQUE=47.7)	~~A06B-1427-B103~~ A06B-1467-B123#0021	C067500076
SPA12/10Ki (15/11 KW,O/P TORQUE=70)	A06B-1428-B103	
SPB3/10Ki (5.5/3.7KW,18-29A,154V-220V)DELTA	A06B-1444-B103	
SPB6/10Ki (7.5/5.5KW,40-56A,116-220V)DELTA	A06B-1445-B103	
SPB8/8Ki (11/7.5KW,36-56A,TORQUE=35.8)		
SPB12/7Ki (15/11KW,TORQUE=52.5Nm)		
CLAMPED AT	8000 RPM	
GEAR RATIO	1:1 1:1.5	1:2 (DIRECT / INDIRECT)
SPINDLE MOTOR ON BY STAR \ DELTA METHOD	YES \ NO	
REACTOR		
MAKE	FANUC PETRONICS LTD	
TYPE OR MODEL	A8IL-0001-0101#C	
	A8IL-0001-0122-02	
	A8IL-0001-0156 (62A)	
	A8IL-0001-0155 (46A)	
	A8IL-0001-0157 (110A)	T06030609
FOR PMC AXIS & 4TH AXIS REACTOR	A81L-0001-0083#3C (24A)	
	A81L-0001-0083#3C-01 (24A)	
DISCHARGE RESISTER FOR PMM AXIS	A06B-6089-H500 (16 OHMS)	
FOR 4TH AXIS	A06B-6130-H402 (30 OHMS)	
TRANSFORMER -1		
MAKE	SYMCOM \ TS \ MARS	
PRIMARY VOLTAGE	415 V 3PHASE 50 HZ AC	
SECONDARY VOLTAGE	200 V 3 PHASE AC	
POWER IN KVA	8 / 10 / 12 / 16 / 18 / 22 KVA	1506/08/06
TRANSFORMER -2 (FOR REG POWER SUPPLY)		
MAKE	TS	
PRIMARY VOLTAGE	415 V 3PHASE 50 HZ AC	
SECONDARY VOLTAGE	415 V 3PHASE 50 HZ AC	
POWER	~~600VA~~ 700VA	2261/09/06
REGULATED POWER SUPPLY		
MAKE	SIEMENS \ ACCESS \ TELEMECANIQUE\OMRON	
INPUT VOLTAGE	415V 3PHASE AC	
OUTPUT VOLTAGE	24 V DC	
OUTPUT CURRENT	20 AMPS / 10 AMPS	
TYPE OR MODEL	6EP1 436-3BA00 \ ABL 7UPS24200	
QUANTITY	01	Batch 0619

MACHINE HISTORY

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

STABILIZER		
MAKE	NEEL \	
TYPE OR MODEL		
INPUT VOLTAGE RANGE		
OUTPUT VOLTAGE		
OUTPUT CURRENT		
POWER		
ISOLATION SWITCH		
MAKE	SALZER \ ABB \ TELEMEC	
TYPE	LB263 \ 243	
RATED CURRENT	40A \ 63A	
AIR COOLER		
MAKE	SUNBEAM APPLIENCES \ADVANCE	
CONTROL VOLTAGE	230 V AC	
RATED CURRENT	2.6 AMPS \ 2.4A \ 2.8A\3.5A	
SET TEMPERATURE	32 DEGREE	
POWER	1000 WATTS \ 500WATTS	
TYPE OR MODEL	CNC 4000E \ CNC 2000EG \ TURBO1000 R33	
REFRIGERANT	R -22\R134\R134A	26/06/859
ITEM	**TYPE NUMBER**	**SERIAL NUMBER**
SURGE SUPPRESSORS		
MAKE	YSL	
TYPE OR MODEL	YSL-RC3-1 \ 022-575VAC	
QUANTITY	08nos	
	3 WIRE OR 4 WIRE	
TERMINAL STATION		
MAKE	WAGO	TB1,TB3,TB5,TB7,TB9,TB10,TB11
		TB2,TB4,TB6,TB8,TB19
TOOL BREAKAGE SENSOR		
MAKE	OMRON HI-PRECISION LSW	
TYPE OR MODEL	D5A-8515	
CONTROL VOLTAGE	24 V DC(100ma MAX)	
QUANTITY		
D TYPE CONNECTOR (DUST COVERS)		
MAKE	OEN (DUST COVERS - ESSEN)	
9 PIN MALE (QTY)		
9 PIN FEMALE (QTY) [RELAY BOARDS]	02nos	X20 X21A X22 X23 X24
	02nos	X25 X26A X21A X22A X29
25 PIN MALE (QTY)	—	I4
25 PIN FEMALE (QTY) [EMG MONITOR]	01	EMG MONITOR,RS-232
37 PIN MALE (QTY)	03nos	I2 I5 I/O1

MACHINE HISTORY

Date:-

NOTE:- ***CHECK THE MACHINE AS PER THE WORK ORDER***

37 PIN FEMALE (QTY)	01	I3 I5A
50 PIN MALE (QTY)	new mcp *	02 I1 O1(PCB SIDE),I1(MOP SID)
50 PIN FEMALE (QTY)	new mcp *	O1(MOP SIDE)
HARTING CONNECTOR		
MAKE	SAINATH \ LAPP\ INDO ELECTRICAL	FROM CABINET (BACK SIDE)
COOLANT TANK	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
HYDRAULIC POWER PACK	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
CHIP CONVEYOR	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
FUME EXTRACTOR	10 \ 16 \ 24 PIN	LOCATION:
OTHERS		
	10 \ 16 \ 24 PIN	LOCATION:
	10 \ 16 \ 24 PIN	LOCATION:

MACHINE HISTORY

RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
CURRENT SENSING RELAY	YSL	CSR-1 \ CSR 3	24V DC	1	Z-AXIS \ 3-~~AXIS~~ BRAKE
SOLID STATE RELAY	ELEC.RELAY	SJDA - 4825 -00	24V DC	2 4 6	MAGZ ARM INDEX TBL
SOLID STATE RELAY (MODULE)	YSL	ACSSC-2P-1	24V DC	1	MAGZ
A.C.RELAY	OMRON	MK2P-I	230 AC 1P	1 + 1	Z-AXIS BRAKE CONTROL + PANEL COOLER
D.C.RELAY	YSL	YSL-1RDU-WRC-WT-1	24V DC	1	Z-AXIS BRAKE CONTROL
CNC ON/OFF PCB	YSL	RP-002-TC-CP-1	24V DC	1	CNC ON/OFF CONTROL
SINGLE RELAY MODULE	YSL	YSL-UMK-1RM DC-CN-FI-2	24V DC	1	
DOUBLE RELAY MODULE	YSL	YSL-UMK-2RM DCCN-FI-2	24V DC	1	APC
I/O PCB MACHINE	YSL	YSL-S26-009-013-REV1-12 / 04	24V DC	1	
I/O PCB MCP	YSL	YSL-S26-009-012-REV1-12 / 04	24V DC	1	
SOLID STATE RELAY MODULE					

MACHINE HISTORY

RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
2-RELAY MODULE	YSL	YSL-SSRP-002-CN-1	24V DC	02nos	01) mag and arm brake 02) star-delta.
4-RELAY MODULE	YSL	YSL-SSRP-004-CN-5	24V DC	—	
8-RELAY MODULE	YSL	YSL-SSRP-008-CN-1	24V DC	01	Hyd fix
RELAY BOARD WITH RELAYS	YSL	S26-010-14-CN	24 V DC	01	X20 X21 X22 X23 X24
SOURCE TYPE		S26-009-23-CN			X25 X26 X27 X28
LAMPS					
THREE TIER LAMP	~~PATANKAR~~	Special with heater	24 V DC	LEDTYPE	01 LAMP TYPE
SINGLE TIER LAMP(GREEN)			24 V DC	LEDTYPE	/ LAMP TYPE
MACHINE LAMP	TECHLITE / ~~RAJDEEP~~/PATHANKAR		230 V AC 1P		2 NOS / 1 NO
CABINET LAMP	PASSOLITE		230V AC 1P		2 NOS / 1 NO

MACHINE HISTORY

M.O.P ELEMENTS	MAKE	TYPE	QTY	PURPOSE
APC MOP			1	
ROTARY SWITCH				
	OEN	2POLE 4WAY (A17403) AM1	04nos	2 nos Tosoku make
	~~OEN~~	4POLE 16WAY (A51200) 0M2	01	Tosoku make
	OEN	5POLE 10WAY (A17402) AM3	01	Tosoku make
	ELCOM	KNOB (OL-102)	06nos	4 nos Tosoku make
SELECTOR SWITCH				
	TEKNIC	WITH KEY	02nos	AUTO DOOR/ SER ARM/ MAG SEL / PRG SEL /SEAT CHECK / DNC
	TEKNIC	2 POSITION	—	DISPSEL CLNTGUN APC STN1 OR STN2
	TEKNIC	2 POSITION WITH SPRING RTN	01	CHIP CONV FWD \ REV
	ELCOM	EDIT KEY SWITCH (AB)	01	PROGRAM EDIT
	TEKNIC	TOGGLE SWITCH		

MACHINE HISTORY

M.O.P ELEMENTS	MAKE	TYPE	QTY	PURPOSE
PUSH BUTTONS				
	TEKNIC	GREEN PUSH BUTTON	05 Nos	CYC ON, CLP, A DOOR, EJCTR UP/DN, CLP 2, LOC FWD/REV,APC ST
	TEKNIC	RED PUSH BUTTON	02 Nos	CYC OFF,DCLP,APC STOP
	TEKNIC	BLACK PUSH BUTTON	—	
	TEKNIC	MUSHROOM PB(RED)	02	MACHINE EMG, CCR EMG, DOUBLE HAND EMG,APC EMG
	TEKNIC	MUSHROOM PB(GREEN)/(BLACK)	02	DOUBLE HAND CYCLE START
	TEKNIC	LUMINIOUS PB(GREEN)	01	TOOL DCLP / PAL SETUP (APC MOP)
	PARA	RECTANGULAR PUSH BUTTON WITH GREEN LENS + NO BLOCK T.H. SERIES		
	PARA	RECTANGULAR PUSH BUTTON WITH RED LENS + NO BLOCK T.H. SERIES		
	PARA	RECTANGULAR PUSH BUTTON WITH RED LENS + NO BLOCK WITH PROTECTIVE COVER T.H.		
	PARA	NON ILL. YELLOW P.B.T.H. MAKE WITH NO BLOCK		

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
PROXIMITY				
SENSORS	TEKNIC	M12 X 73 MM CONN TYPE	08	A
	TEKNIC	M12 X 52 MM CONN TYPE	—	B
	TEKNIC	M12 X 40 MM NON CONN TYP	—	C
	TEKNIC	M12 X 68 MM NON CONN TYP	—	D
	TEKNIC	M8 X 50 MM NON CONN TYP	04	E
	YSL	M12 X 50 MM NON CONN TYP(PNP)	—	G
	TEKNIC	EGT12X04UP 024-2000L	—	F
	TEKNIC	M8 CONN TYPE	—	H
	TEKNIC	CABLE(SWLF3AP-5000P)	Olny	
	TEKNIC	CABLE(SWLF3AP-2000P)	—	
		SPINDLE ORIENTATION.PXS	01	A B C D E G F H
		TOOL CLAMP PROXIMITY	01	A B C D E G F H
		TOOL DECLAMP PROXIMITY	01	A B C D E G F H
		ONE ROT.SIGNAL PROXIMITY	—	A B C D E G F H
		MAG IN POSITION PROXIMITY	—	A B C D E G F H
		NO TOOL IN SPINDLE PROXIMITY	01	A B C D E G F H
		COOLANT LEVEL LOW	—	A B C D E G F H
		CHIP CONVEYOR O/L PROXIMITY	—	A B C D E G F H
		PROXIMITIES ON APC TABLE	—	A B C D E G F H
		PROXIMITIES ON APC STATION	—	A B C D E G F H
		CHIP CONVEYOR O/L PROXIMITY	—	A B C D E G F H
LIMIT SWITCH				
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	M/C DOOR CLOSE LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	CABINET DOOR CLOSE LIMIT SWT
	TEKNIC\BCH	NLSPR \ JAI BALAJI	—	PALLET-1 STATION LIMIT SWITCH
	TEKNIC\BCH	NLSPR \ JAI BALAJI	—	PALLET-2 STATION LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	01	CHIP CONV1 O/L LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	CHIP CONV2 O/L LIMIT SWITCH

MACHINE HISTORY

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
	BCH	NLSPR \ JAI BALAJI	—	APC DOOR UP LIMIT SWITCH
	BCH	NLSPR \ JAI BALAJI	—	APC DOOR DOWN LIMIT SWITCH
	TELEMECHA	XCE-154 \ JAI BALAJI	—	CYCLE START LIMIT SWITCH
	TEKNIC	NIA K502	—	FIXTURE LIMIT SWITCH
	TEKNIC	SG-01 R552	—	FIXTURE LIMIT SWITCH
REED SWITCHES				
	FESTO		06 nos	3 Auto door MACHINE DOOR CLOSE / ~~FIXTURE~~
SENSORS				
FLAT SENSOR	TEKNIC	FG2X05APO24-5000L	1	MAG REF POSITION
FLOAT SENSOR (FALLING)		OLG-24L	01	DIRTY COOLANT LEVEL LOW
FLOAT SENSOR (RAISING)			—	DIRTY COOLANT LEVEL HIGH
FLOAT SENSOR (FALLING)			—	CLEAN COOLANT LEVEL LOW
FLOAT SENSOR (RAISING)				CLEAN COOLANT LEVEL HIGH
MCB'S				
PROTEC / MERLIN GERIN				
F15		25A / 3P 40 / 3P 63A / 3P	01	TRANSFORMER PRIMERY
F16		3A/3P	01	REG. POWER SUPPLY
F17		30mA, 63A	—	RCCB FOR PWR CIRCUIT
F18		10A/1P	01	230V AC SUPPLY
F19		40A/3P 63A/3P 100/3P	01	TRANSFORMER SECONDARY
F20		1A/3P 3A/3P 5A/3P	01	SPINDLE FAN / CP1A

MACHINE HISTORY

MAKE : SPRECHER & SCHUH / KLOCKNER MOELLER

MPCB NO.	CONT. NO	MPCB RATING	MPCB SETTING	DESCRIPTION	MAKE	MODEL	RATING	REMARKS
F1	K1 & K2	1-1.6A	1.6A	MAGZINE MOTOR	PRAGATI	TORQUE MTR	0.9KVA	
		2.5-4A	2.54A ✓				1.9KVA ✓	
F2	K3	2.5-4A ✓	2.5A	ARM MOTOR	PRAGATI	TORQUE MTR	1.6KVA	
			3.2A ✓				2.3KVA ✓	
F3	K4	0.63-1A	0.54A	LUB MTR	H.MTR MFG	INDUCTION MTR	0.09KW ✓	
		1A/3P	MCB ✓					
F4	K5	1-1.6A	1.08A	FLOOD CLNT MTR	RAJMANE	RK3-270	0.37KW	
		4-6.3A	5.4A ✓		RAJMANE	RKM-206	2.25KW ✓	
					GRUNDFOS		2.25KW	
F5	K6	1-1.6A	1.08A	CHIP PUSH CLNT MTR	RAJMANE	RK3-270	0.37KW	
		4-6.3A	5.4A ✓		RAJMANE	RKM-206	2.25KW ✓	

MACHINE HISTORY

MPCB NO.	CONT. NO	MPCB RATING	MPCB SETTING	DESCRIPTION	MAKE	MODEL	RATING	REMARKS
F6	K7	1-1.6A	1.08A	CHIP FLUSH CLNT MTR	RAJMANE	RK3-270	0.37KW	
		4.6.3A	5.4A		RAJMANE	RKM-206	2.25KW	
F7	K8	1-1.6A	1.08A	CHIP PUSH ON CHIP	RAJMANE	RK3-270	0.37KW	
				CONVEYOR/CLNT GUN				
F8	K9	2.5-4A	3A	HYDRAULIC MTR	SIEMENS		1.1KW	
		4-6.3A						
F9	K10	4-6.3A	6.18A	H.P.CLNT MTR	GRUNDFOS		2.2KW	
		6-10A			B.BIJLEE		3.7KW	
		10-16A			WANNER			
F10	K11 & K12	0.63-1A	0.78A	CHIP CONV.-1 MTR	LHPL / BONFIGLOLI		0.18KW/	(SLAT/SCRAPER)
		1-1.6A	1.2A		BONFIGLOLI		0.37KW	
F11	K13 & K14	0.63-1A	0.78A	CHIP CONV.-2 MTR	LHPL / BONFIGLOLI		0.18KW/	(SLAT/SCRAPE)
		1-1.6A	1.2A		BONFIGLOLI		0.37KW	
F12	K15 & K16	2.5-4A	2.76A	INDEX TABLE MTR	PRAGATI		1.6 KVA	

MACHINE HISTORY

MPCB NO.	CONT. NO	MPCB RATING	MPCB SETTING	DESCRIPTION	MAKE	MODEL	RATING	REMARKS
F13	K17 & K18	0.63-1A	0.78A	SCREW CONV.MTR	LHPL / BONFIGLOLI		0.18KW/	
		1-1.6A	1.2A		BONFIGLOLI		0.37KW	
F14	-	0.01-0.16A		24VA TFR INPUT	-	-	-	
F21	K23	0.10-0.16A		OIL SKIMMER OPTION				
F28	K26	2.5-4A		FUME EXTRACTOR				
F27	K25	1-1.6A	1.08A	COOLANT GUN MTR	RAJMANE	RK3-270	0.37KW	
F33	K30	4-6.3A	5.4A	BAG FILTER MTR	RAJMANE	RKM-206	2.25KW	

CONTACTORS	TYPE	MAKE	QTY	PURPOSE
K	CA4-9C-01 24V DC	SPRECHER SHUH / A.B	16nos	
PSM CONTACTOR	CA3-30C-01	SPRECHER SHUH	1	PSM
	CA3-43C-01	SPRECHER SHUH		
	CA3-43-N-11	SPRECHER SHUH	1	PSM
	CA3-72-N-11	SPRECHER SHUH	3	STAR/DELTA
ADD ON BLOCK		AB / SPRECHER SHUH		SD / HYD C.FAN
EMG RELEASE	CS4C-40E	SPRECHER SHUH / AB	1	EMG CONTR

		YES/NO	REMARKS
	ROTARY TABLE\ INDEX TABLE		
1	INTERFACING OF ROTARY TABLE	—	
2	INTERFACING OFINDEX TABLE	—	
3	ROTARY TABLE MAKE [NIKKEN]	—	
4	ROTARY TABLE MAKE [UCAM]	—	
5	INDEX TABLE STATIONS [8 \12 \24]	—	
6	INDEX TABLE CABLE LENGTH IN MTS	—	
7	ROTARY TABLE CABLE LENGTH IN MTS	—	
8	DRIVES FOR ROTARY TABLE [SVM BETA OR SVM]	—	
9	LOCATION OF MOUNTING OF JUNCTION BOX / CABINET [BOTTOM PLATE LH OR RH FROM FRONT]	—	
10	SERVO TURRET OPTION		
		—	
	FIXTURE		
1	PNEUMATIC FIXTURE (NO.OF STATIONS)	—	
2	HYDRAULIC FIXTURE (NO.OF STATIONS)	YES	
3	FIXTURE M .O. P OPTION	YES	
4	COMP CLAMP OPTION WITH [1 / 2]PB IN MOP	YES	
5	COMP DECLAMP OPTION WITH [1 / 2]PB IN MOP	YES	
6	COMP CLAMP PR. SW OPTION	YES	
7	COMP CLAMP AND DECLAMP PB(COMMON PB)	—	
8	EJECTOR UP/DOWN COMMON PB	—	
9	EJECTOR PRSW OPTION	—	

		YES/NO	REMARKS
10	COMP ORIENTATION PR. SW.OPTION	—	
11	COMP ORIENT AND DEORIENT PB(COMMON PB)	—	
12	COMP JACK LOCK OPTION WITH [1 / 2] PB IN MOP	—	
13	COMP JACK UNLOCK OPTION [1 / 2] PB IN MOP	—	
14	COMP JACK LOCK PR.SW OPTION	—	
15	JACK LOCK AND UNLOCK PB(COMMON PB)	—	
16	TAIL STOCK FWD OPTION WITH [1 / 2] PB IN MOP	—	
17	TAIL STOCK REV OPTION WITH [1 / 2] PB IN MOP	—	
18	TAIL STOCK FWD PRESSURE	—	
19	TAIL STOCK FWD AND REV PB(COMMON PB)	—	
20	COMP REST SENSE OPTION	—	
21	NO.OF REST SENSE SOLENOIDS [1 OR 2]	—	
22	REST SENSE PRESSURE SWITCHES [1 2 3 4]	—	LOW/HIGH
23	ANY OTHER SPECIFY		
23A	Low prs work support up & down	YES	
23B	High prs work support up & down	YES	
24	FIXTURE SEQUENCE TO BE ATTACHED	—	
25	AUTO DOOR [Front, LHS, RHS]	YES/NO	

	OTHERS	YES/NO	REMARKS
1	AIR PRESSURE SWITCH + PNEUMATIC COUNTER BAL	YES	
2	MANUAL PALLET CHANGER	—	
3	HAND HELD MPG	YES	
4	SEALED KEY BOARD ~~MCP~~	YES	new with Tosoku sw.
5	Z-AXIS BRAKE	YES	
6	X & Y-AXIS BRAKE	—	
7	OIL SKIMMER OPTION	—	
8	OIL COOLER OPTION	—	
9	DHPB OR SINGLE LEVER CYCLE START SWITCH	YES	
10	MIST EXTRACTOR OPTION	—	
11	SPINDLE LOAD METER (LM5)	YES	
12	TOOL STATION DISPLAY UNIT (LM5)	YES	
13	RENISHAW / MONTRONIX / M & H PROBE	—	
14	SEPARATE COOLANT GUN MTR	—	
	SERVO ATC	—	
2	SERVO ARM OPTION	—	
3	SERVO MAGAZINE OPTION	—	
4	NO. OF POCKETS IN MAGAZINE	YES	30 tool.
	APC	—	
1	LENGTH FOR APC PROXIMITY ON TABLE	—	
2	TYPE OF LIMIT SWITCH FOR PALLET ON STATION	—	STRAIGHT / 90DEG
3	TYPE OF LIMIT SWITCH FOR DOOR UP AND DOWN	—	STRAIGHT / 90DEG
		YES/NO	REMARKS

	AXES		
1	STRAIGHT OR BENT CONNECTORS FOR AXIS POWER	YES	X-axis bent type
2	a)DROP PREVENTER UNIT - CAPACITOR MODULE	YSL	
	b)POWER FAILURE DETECTION UNIT	SUCHITHRA	YES 066
	CONFIRM FOLLOWING DATA		
1	KEEPRELAY AND TIMER DATA	YES	
2	PARAMETERS ,PMC PARAMETERS,PITCH ERROR DATA	YES	
3	LADDER	YES	
4	SRAM DATA BACKUP	YES	
5	FUNCTIONAL CHECKING	YES	
6	FIXTURE I/O'S, TB DETAILS ETC TO BE ATTACHED		
6	PUT BACKUP IN PC AND ENTER IN PENDING LIST		

	SPECIAL INFORMATIONS
1	
2	
3	
4	
5	
6	

SIGNATURE:

NAME: Prakash Bhat

DATE: 03/10/06